

PH101: 2023

Special Theory of Relativity

Lecture-5

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Recap

- After relativistic kinematics, position, time, we enter into the realistic dynamics
- We defined relativistic mass, which keeps momentum conserved in all frames

$$m = \frac{m_0}{\sqrt{1 - u^2/c^2}}$$

- We defined relativistic momentum, following relativistic mass as:

$$\mathbf{p} = \frac{m_0 \mathbf{u}}{\sqrt{1 - u^2/c^2}} \rightarrow \vec{p} = \gamma m_0 \vec{u}$$

- Mass-energy equivalence relation, which in turn provides a relativistic expression for kinetic energy:

$$E = mc^2$$

$$E = m_0 c^2 + K$$

$$E^2 = p^2 c^2 + m_0^2 c^4$$

Force and Acceleration

Force:

$$\mathbf{F} = d\mathbf{p}/dt = \frac{d}{dt}(m\mathbf{u}) = m \frac{d\mathbf{u}}{dt} + \mathbf{u} \frac{dm}{dt}$$

$$E = mc^2 \quad m = E/c^2$$

$$\frac{dm}{dt} = \frac{1}{c^2} \frac{dE}{dt} = \frac{1}{c^2} \frac{d}{dt} (K + m_0 c^2) = \frac{1}{c^2} \frac{dK}{dt}$$

Transformation of acceleration to another inertial frame is a bit complicated and we will avoid the discussion here.

$$\frac{dK}{dt} = \frac{(\mathbf{F} \cdot d\mathbf{l})}{dt} = \mathbf{F} \cdot \frac{d\mathbf{l}}{dt} = \mathbf{F} \cdot \mathbf{u}$$

$$\frac{dm}{dt} = \frac{1}{c^2} \mathbf{F} \cdot \mathbf{u}.$$

$$\mathbf{F} = m \frac{d\mathbf{u}}{dt} + \frac{\mathbf{u}(\mathbf{F} \cdot \mathbf{u})}{c^2}$$

$$\mathbf{a} = \frac{d\mathbf{u}}{dt} = \frac{\mathbf{F}}{m} - \frac{\mathbf{u}}{mc^2} (\mathbf{F} \cdot \mathbf{u})$$

In general, the acceleration is in the same direction as of the force, but in relativity, it is not, due to the presence of a term proportional to the velocity.

Compton scattering (1919-1923)

Compton showed that the radiation of a X-ray of given wavelength is not consistent with classical radiation theory

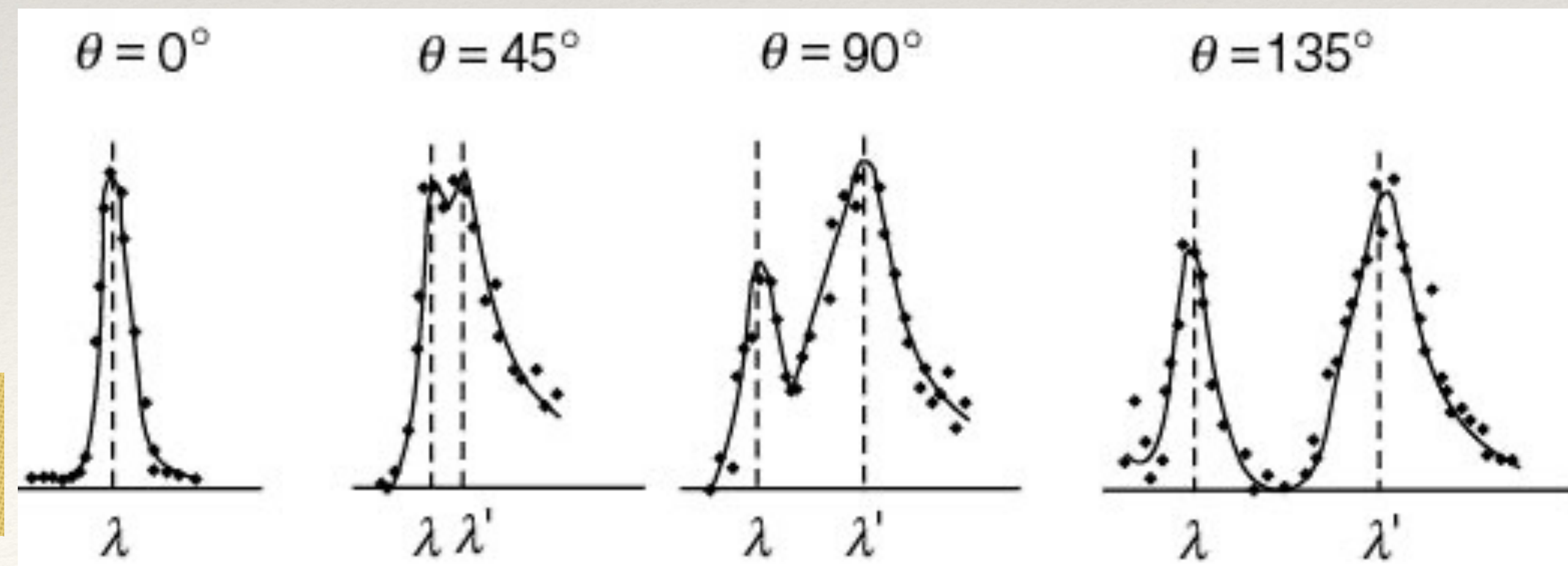
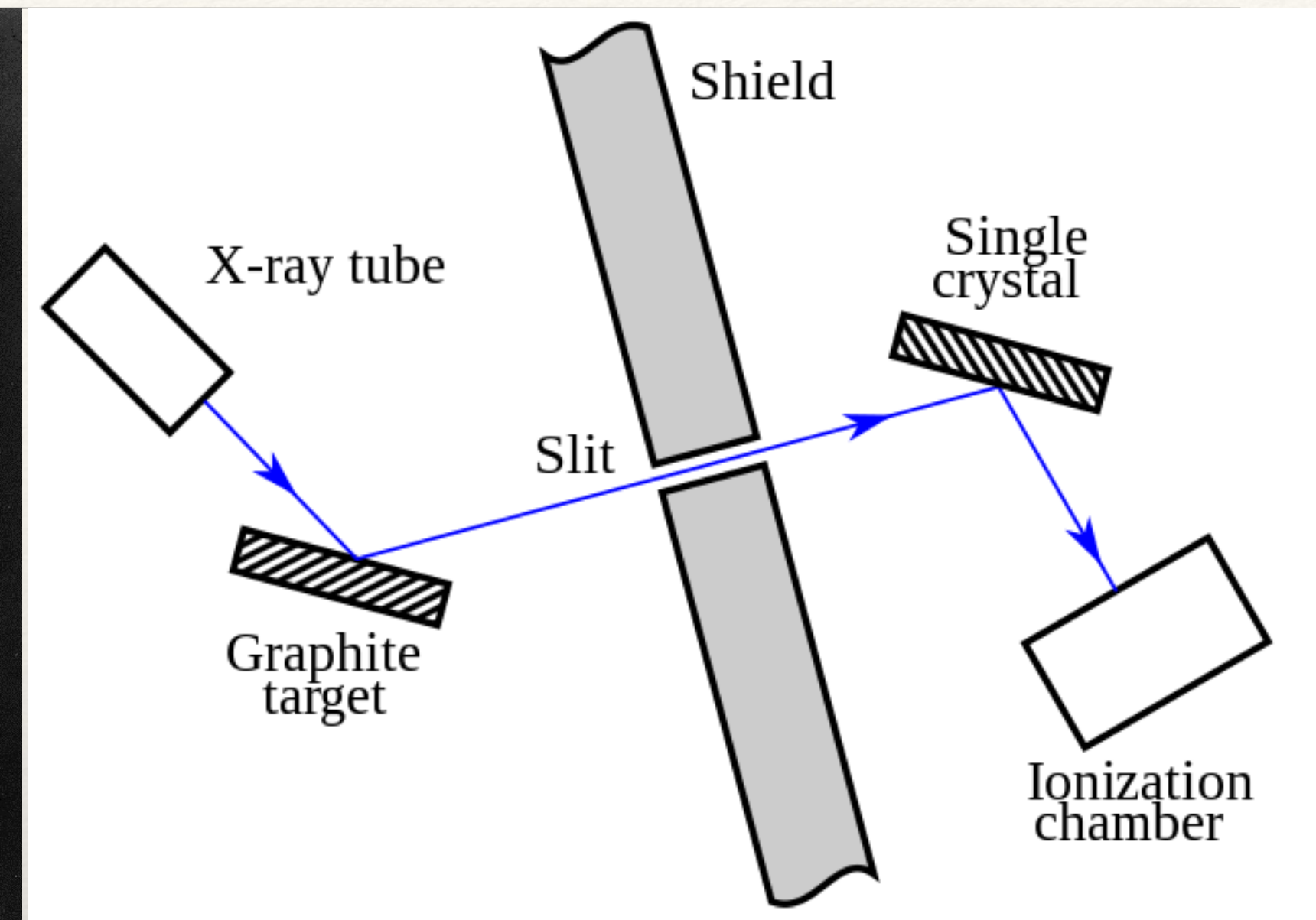
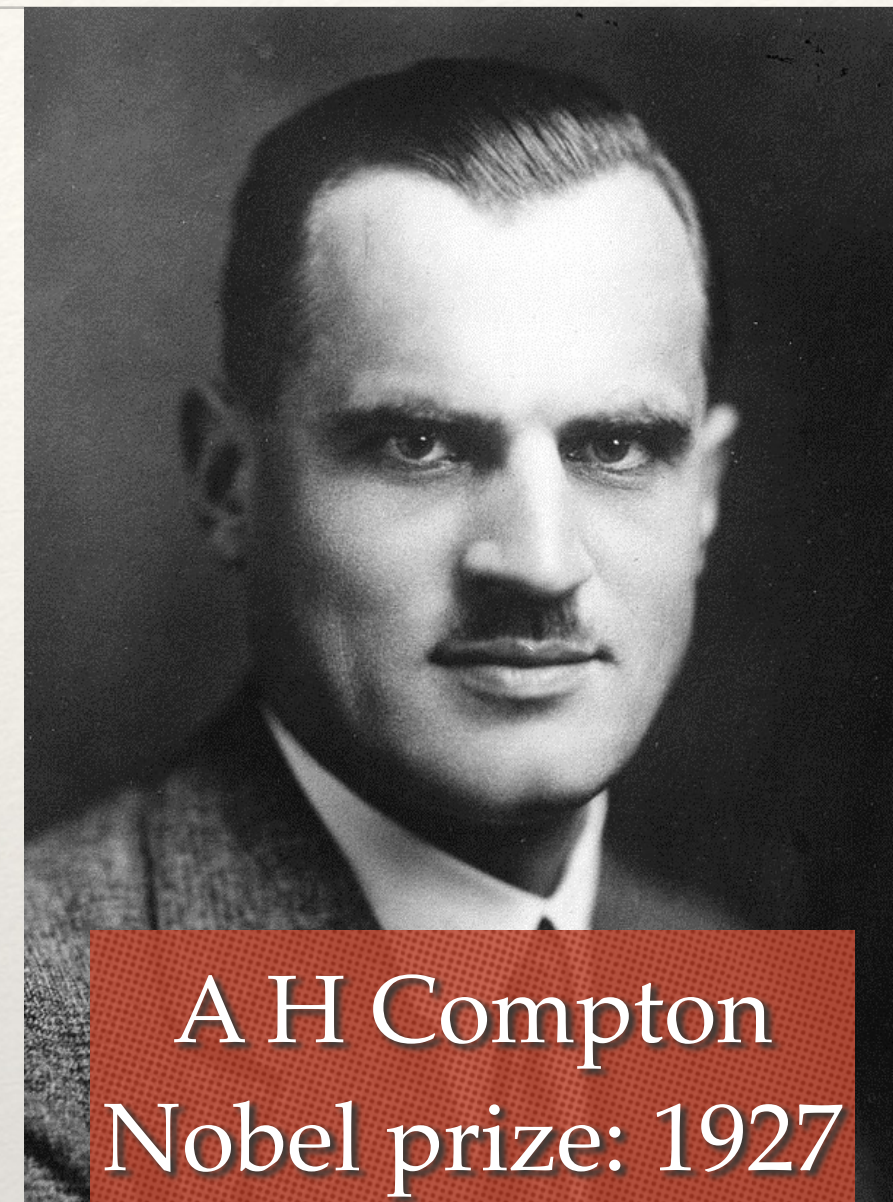
- Classical theory:
- There is no way to generate a new wavelength
 - The intensity doesn't depend on the wavelength of incident radiation

Observation:

The radiation scattered through a given angle consists of two components:

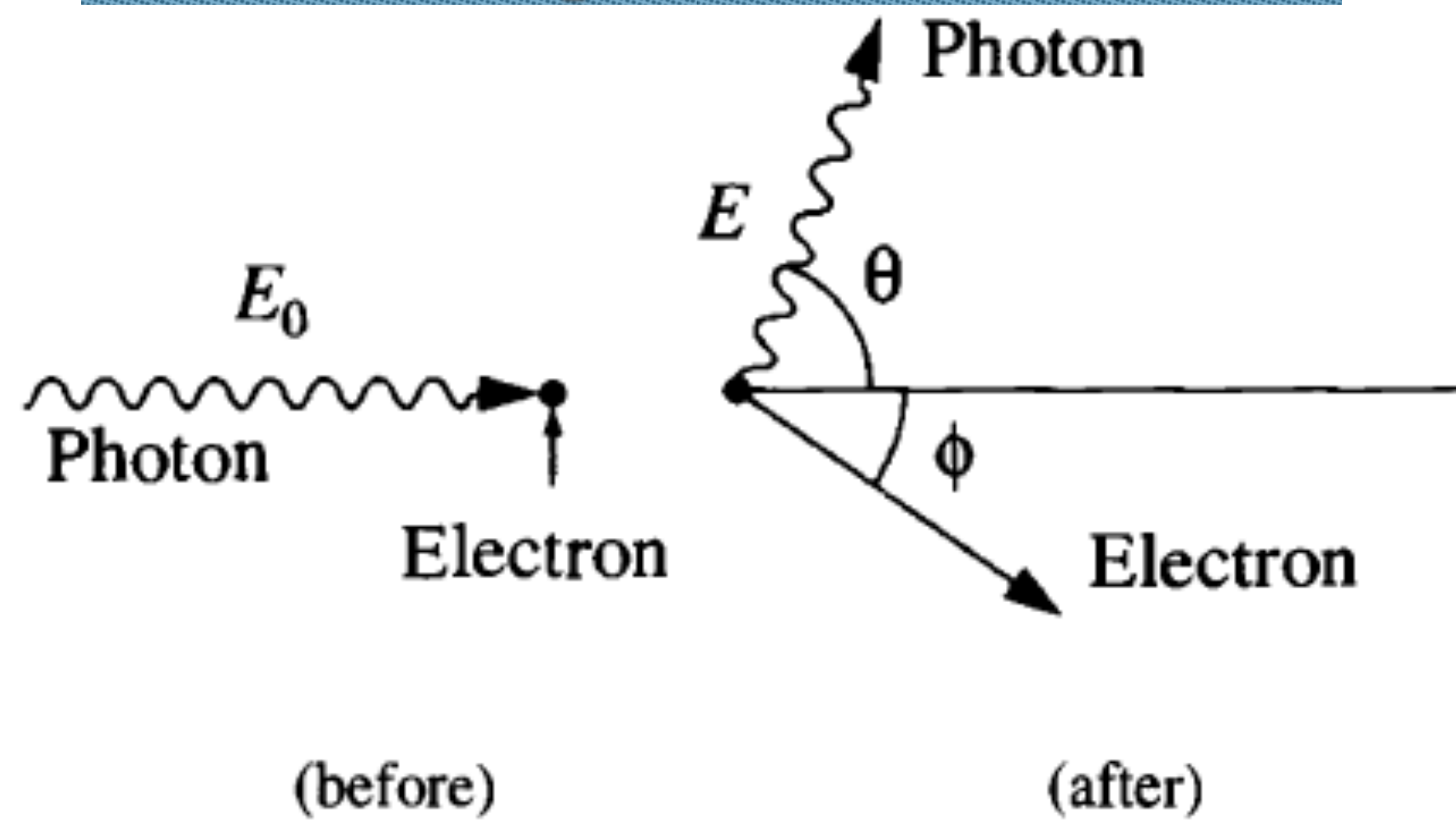
- One, whose wavelength is the same as the original one,
- The other wavelength shifted relative to the incident wavelength by an amount depending on the angle.**

The modified wavelength could be described by assuming the radiation as beam of photons having energy $E = h\nu$.



Compton scattering

Collision of photon and electron



- Conservation of momentum in the *vertical* direction gives

$$p_e \sin \phi = p_p \sin \theta, \quad \sin \phi = \frac{E}{p_e c} \sin \theta.$$

- Conservation of momentum in the *horizontal* direction gives

$$\frac{E_0}{c} = p_p \cos \theta + p_e \cos \phi = \frac{E}{c} \cos \theta + p_e \sqrt{1 - \left(\frac{E}{p_e c} \sin \theta \right)^2}$$

This couldn't be explained using classical energy-momentum Relations

$$p_e^2 c^2 = (E_0 - E \cos \theta)^2 + E^2 \sin^2 \theta = E_0^2 - 2E_0 E \cos \theta + E^2.$$

- conservation of energy: $E_0 + mc^2 = E + E_e = E + \sqrt{m^2 c^4 + p_e^2 c^2}$
 $= E + \sqrt{m^2 c^4 + E_0^2 - 2E_0 E \cos \theta + E^2}.$

$$E = \frac{1}{(1 - \cos \theta) / mc^2 + (1/E_0)}.$$

Using,

$$E = h\nu = \frac{hc}{\lambda},$$

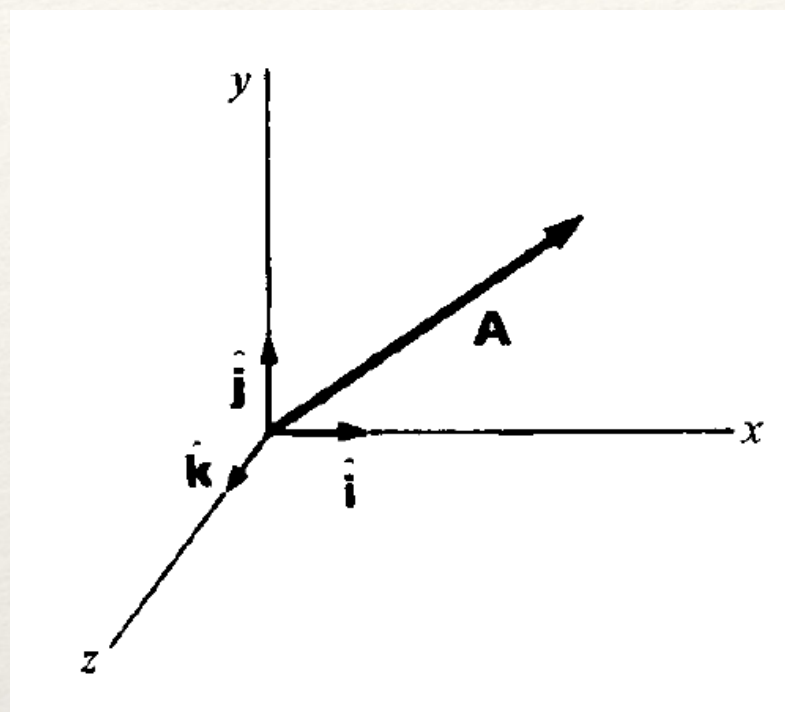
We find,

$$\lambda = \lambda_0 + \left(\frac{h}{mc} \right) (1 - \cos \theta).$$

$$\Delta \lambda = \frac{h}{mc} (1 - \cos \theta) \quad \text{Explains the observation!}$$

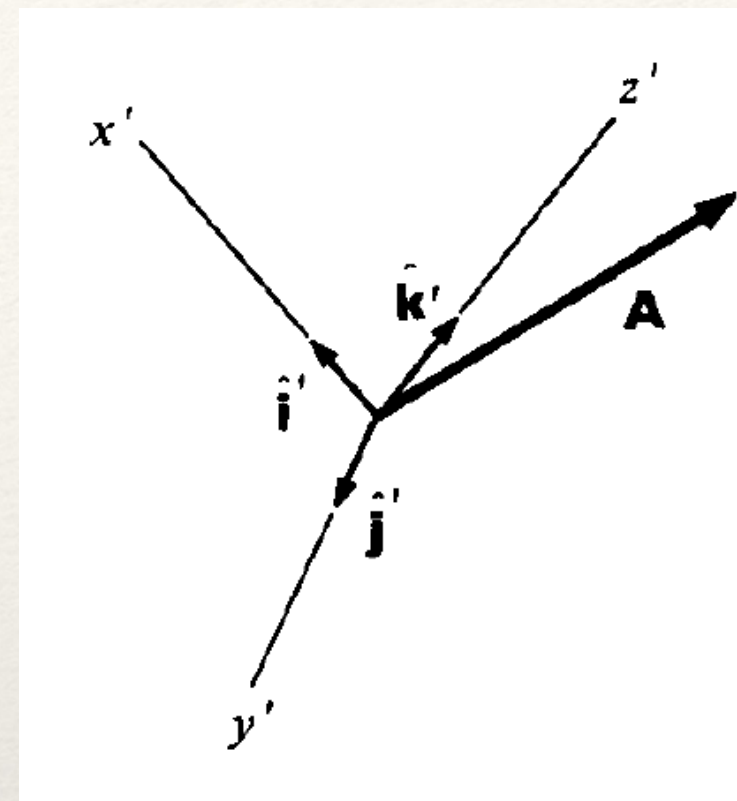
Compton wavelength of electron

Vectors and transformations



A vector in 3 D

$$\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}.$$



The same vector in different coordinate system:

$$\mathbf{A} = A'_x \hat{\mathbf{i}}' + A'_y \hat{\mathbf{j}}' + A'_z \hat{\mathbf{k}}'.$$

Both are same !!

$$A'_x \hat{\mathbf{i}}' + A'_y \hat{\mathbf{j}}' + A'_z \hat{\mathbf{k}}' = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}.$$

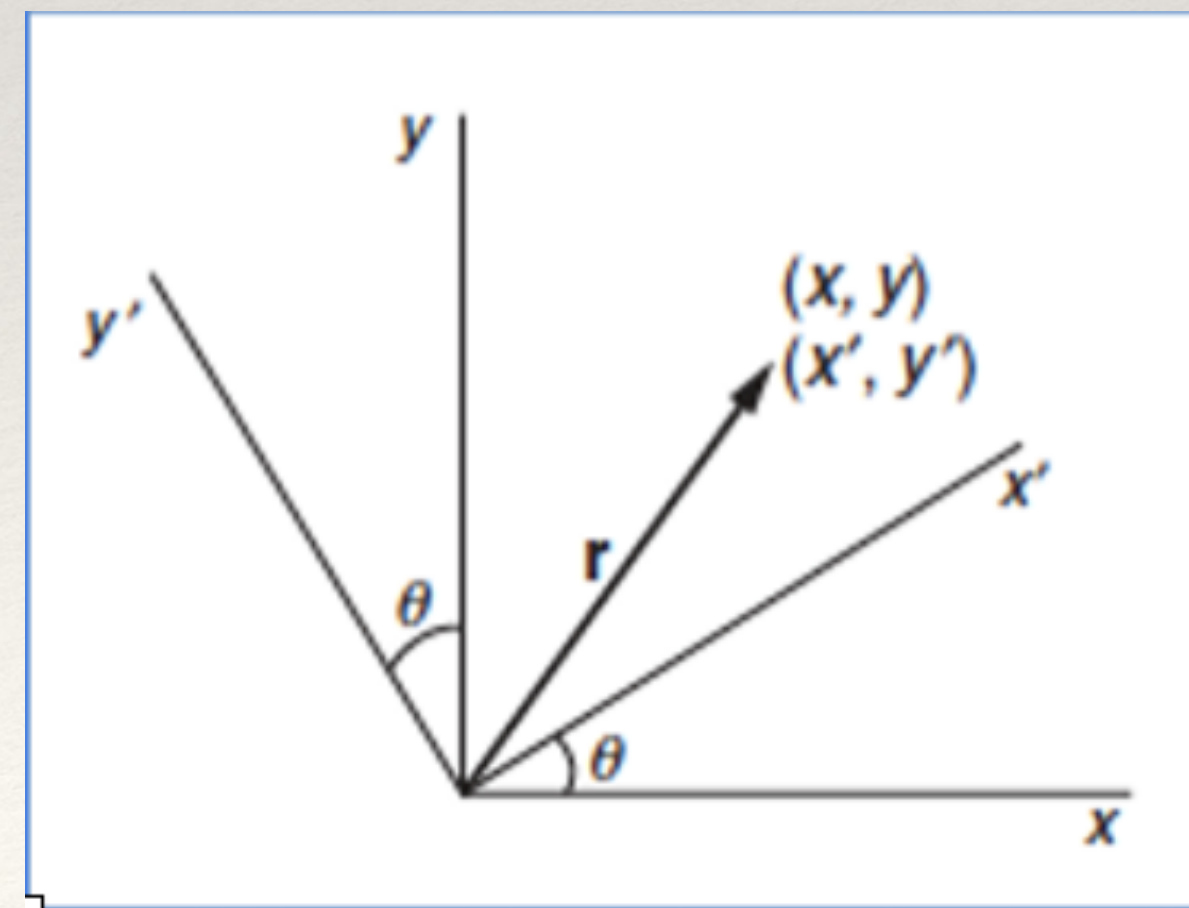
Take a dot product with $\hat{\mathbf{i}}'$

$$A'_x = A_x(\hat{\mathbf{i}}' \cdot \hat{\mathbf{i}}) + A_y(\hat{\mathbf{i}}' \cdot \hat{\mathbf{j}}) + A_z(\hat{\mathbf{i}}' \cdot \hat{\mathbf{k}})$$

Similarly,

$$\begin{aligned} A'_y &= A_x(\hat{\mathbf{j}}' \cdot \hat{\mathbf{i}}) + A_y(\hat{\mathbf{j}}' \cdot \hat{\mathbf{j}}) + A_z(\hat{\mathbf{j}}' \cdot \hat{\mathbf{k}}) \\ A'_z &= A_x(\hat{\mathbf{k}}' \cdot \hat{\mathbf{i}}) + A_y(\hat{\mathbf{k}}' \cdot \hat{\mathbf{j}}) + A_z(\hat{\mathbf{k}}' \cdot \hat{\mathbf{k}}). \end{aligned}$$

Example: Rotation



$$\begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= -x \sin \theta + y \cos \theta \\ z' &= z \end{aligned}$$

$$\begin{aligned} \hat{\mathbf{i}}' &= \hat{\mathbf{i}} \cos \theta + \hat{\mathbf{j}} \sin \theta, \\ \hat{\mathbf{j}}' &= -\hat{\mathbf{i}} \sin \theta + \hat{\mathbf{j}} \cos \theta, \\ \hat{\mathbf{k}}' &= \hat{\mathbf{k}} \end{aligned}$$

$$\begin{aligned} A'_x &= A_x \cos \theta + A_y \sin \theta \\ A'_y &= -A_x \sin \theta + A_y \cos \theta \\ A'_z &= A_z \end{aligned}$$

Invariants under transformation

Any quantity which remains unchanged under coordinate transformation is called invariant under transformation.

Check the dot product of vectors:

$$\begin{aligned} A'_x B'_x + A'_y B'_y + A'_z B'_z &= (A_x \cos \Phi + A_y \sin \Phi)(B_x \cos \Phi + B_y \sin \Phi) \\ &+ (-A_x \sin \Phi + A_y \cos \Phi) \\ &(-B_x \sin \Phi + B_y \cos \Phi) + (A_z B_z) \\ &= A_x B_x + A_y B_y + A_z B_z \end{aligned}$$

Invariant

Similarly, the **norm of a vector**, the dot product with itself will also remain invariant under the coordinate transformation

$$A'^2_x + A'^2_y + A'^2_z = A^2_x + A^2_y + A^2_z.$$

Now, consider Lorentz transformations:

They are very similar to the spatial transformation like rotation as shown above.

Question: What remains invariant under Lorentz transformation ?

Lorentz's Transformations

$$\begin{aligned} x' &= \frac{x - vt}{\sqrt{1 - v^2/c^2}} \\ y' &= y \\ z' &= z \\ t' &= \frac{t - (v/c^2)x}{\sqrt{1 - v^2/c^2}} \end{aligned}$$

Inverse Lorentz's Transformations

$$\begin{aligned} x &= \frac{x' + vt'}{\sqrt{1 - v^2/c^2}} \\ y &= y' \\ z &= z' \\ t &= \frac{t' + (v/c^2)x'}{\sqrt{1 - v^2/c^2}} \end{aligned}$$

Lorentz Invariance

Locus of a light pulse : $x^2 + y^2 + z^2 = (ct)^2$

The same in a frame moving with velocity v is given by:

$$x'^2 + y'^2 + z'^2 = (ct')^2.$$

comparing both we see that $x^2 + y^2 + z^2 - (ct)^2$ is zero in both coordinate system and it appears to be invariant under Lorentz transformation.

Check:

$$x'^2 + y'^2 + z'^2 - (ct')^2 = \gamma^2(x - vt)^2 + y^2 + z^2 - \gamma^2 c^2 \left(t - \frac{vx}{c^2}\right)^2$$

$$= \frac{1}{1 - v^2/c^2} \left[x^2 \left(1 - \frac{v^2}{c^2}\right) - c^2 t^2 \left(1 - \frac{v^2}{c^2}\right) \right] + y^2 + z^2$$

$$= x^2 + y^2 + z^2 - (ct)^2.$$

$x^2 + y^2 + z^2 - (ct)^2$ remains invariant under Lorentz transformation.

$x^2 + y^2 + z^2$ remains invariant under rotation in 3 dimension

Minkowski
Space



Hermann Minkowski
(1864-1909)

Construct a 4 Dimensional space with components

$$x_1 = x, x_2 = y, x_3 = z, x_4 = ict (i^2 = -1)$$

Thus,

$$||\mathbf{x}|| = \sum x_i x_i = (x_1^2 + x_2^2 + x_3^2 + x_4^2)$$

Remains Lorentz invariant

Four vectors

Minkowski Space:

$$\mathbf{S} = \{\vec{r}, ict\}$$

We can therefore
define four vectors

$$\mathbf{A} = \{A_i\}, i = 1 - 4$$

Under Lorentz
transformation

$$A'_1 = \gamma(A_1 + i\beta A_4)$$

$$A'_2 = A_2$$

$$A'_3 = A_3$$

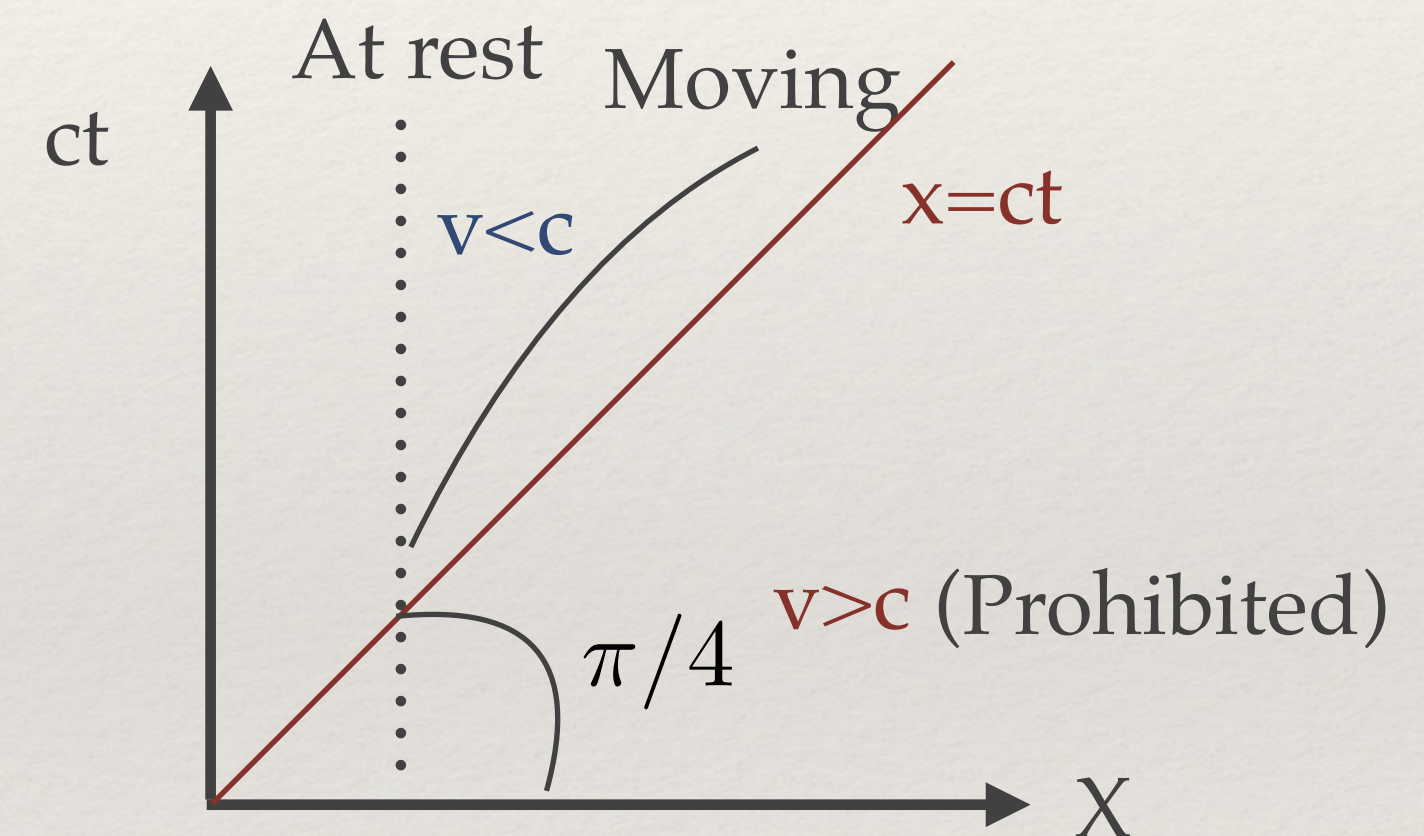
$$A'_4 = \gamma(A_4 - i\beta A_1)$$

$$\beta = v/c$$

Transformations
are similar to
Minkowski
space time
vectors

A point in the **Minkowski space** is called a world point

A particle's movement in space time points
yield a **world line**.



The norm of $\mathbf{A}$ remains invariant under Lorentz transformation:

$$A_1'^2 + A_2'^2 + A_3'^2 + A_4'^2 = A_1^2 + A_2^2 + A_3^2 + A_4^2.$$

Four vectors and proper time

World point: $\mathbf{S} = \{\vec{r}, ict\}$

$$\Rightarrow d\mathbf{S} = \{d\vec{r}, icdt\}$$

Lorentz invariant quantity: $d\mathbf{S}^2 = dx^2 + dy^2 + dz^2 - c^2 dt^2$

Define:

$$\Rightarrow d\tau^2 = -\frac{1}{c^2} d\mathbf{S}^2 = dt^2 - \frac{1}{c^2} (dx^2 + dy^2 + dz^2)$$

$$= \underbrace{dt'^2 - \frac{1}{c^2} (dx'^2 + dy'^2 + dz'^2)}_{=0}$$

Proper Time (Time interval that is Lorentz invariant)

In the rest frame of the particle, $dx' = dy' = dz' = 0 \Rightarrow d\tau = dt'$

Time interval measured in rest frame

Consider an observer at rest in the x', y', z', t' system.

the proper time interval between two world points is $d\tau = dt'$.

Since $d\tau^2$ is a Lorentz invariant, its value for the same world points is the same in all frames. Hence, we can equate its value in the rest frame to its value in the second frame.

$$dt'^2 = dt^2 - \frac{1}{c^2} (dx^2 + dy^2 + dz^2)$$

$$\left(\frac{dt'}{dt}\right)^2 = 1 - \frac{1}{c^2} \left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2 \right]$$

$$\left(\frac{dt'}{dt}\right)^2 = 1 - \frac{v^2}{c^2}$$

Using,

$$d\tau = dt'$$

$$dt = \gamma d\tau$$

Time dilation

Four velocity and four momentum

$$d\mathbf{S} = \{d\vec{r}, icdt\}$$

Infinitesimal displacement
along World line

Define, four velocity:

$$\mathbf{u} = \frac{d\mathbf{S}}{d\tau}$$

$$\frac{d\mathbf{S}}{d\tau} = \left\{ \frac{d\vec{r}}{d\tau}, ic \frac{dt}{d\tau} \right\}$$

Proper time

$$\Rightarrow \mathbf{u} = \gamma \{ \vec{u}, ic \}$$

$$\text{As, } dt = \gamma d\tau$$

$\mathbf{u}$ = Relativistic four-velocity, $\vec{u}$ = Newtonian three-velocity

Check: $\mathbf{u}^2 = \gamma^2 (u^2 - c^2)$
 $= -c^2$

Lorentz
invariant

Define, four momentum with
respect to four velocity

$$\begin{aligned} \mathbf{P} &= m_0 \mathbf{u} = \gamma \{ m_0 \vec{u}, im_0 c \} \\ &= \{ m \vec{u}, imc \} \\ &= \{ \vec{p}, imc \} \\ &= \{ \vec{p}, iE/c \} \end{aligned}$$

$$E = mc^2$$

Check:

$$\begin{aligned} \mathbf{P}^2 &= p^2 - \frac{E^2}{c^2} = m_0^2 \gamma^2 (u^2 - c^2) \\ &= -m_0^2 c^2 \end{aligned}$$

Lorentz
invariant

Rest mass energy

$$E^2 = p^2 c^2 + (m_0 c^2)^2,$$

Example: Pair production of particles

Creation of electron-positron pair from collision of electron pair

$$e^- + e^- \rightarrow e^- + e^- + (e^- + e^+)$$

What is the threshold energy for this process?

Use conservation of four momentum energy

$$\mathbf{P}_1 + \mathbf{P}_2 = \mathbf{P}_3 + \mathbf{P}_4 + \mathbf{P}_5 + \mathbf{P}_6 \quad (\text{Frame specific})$$

$$(\mathbf{P}_1 + \mathbf{P}_2)^2 = (\mathbf{P}_3 + \mathbf{P}_4 + \mathbf{P}_5 + \mathbf{P}_6)^2$$

(Frame independent
as Lorentz invariant)

$$\mathbf{P}_1 = \left\{ \vec{p}_1, i \frac{E_1}{c} \right\} \quad \mathbf{P}_2 = \{0, i m_0 c\}$$

In the laboratory frame

$$(\mathbf{P}_1 + \mathbf{P}_2)^2 = -2(m_0 c)^2 - 2m_0 E_1$$

Using

$$\mathbf{P}^2 = -m_0^2 c^2$$

$$(\mathbf{P}_3 + \mathbf{P}_4 + \mathbf{P}_5 + \mathbf{P}_6)^2 = \{ \underbrace{\vec{p}_3 + \vec{p}_4 + \vec{p}_5 + \vec{p}_6}_{=0}, 4i m_0 c \}^2 = -16(m_0 c)^2$$

= 0 (In the centre-of-mass frame)

At the threshold, all the particles are produced at rest in the centre-of-mass frame.

Therefore,

$$\underbrace{-2(m_0 c)^2 - 2m_0 E_1}_{\text{Laboratory frame}} = \underbrace{-16(m_0 c)^2}_{\text{Centre-of-mass frame}}$$

$$E_{th} = E_1 = 7m_0 c^2 = 3.57 \text{ MeV}$$

Only the four momentum squared can be equated in two different frames.

Grand Summary

Now we know Special Theory of Relativity !

But we don't know General Theory of Relativity ! Never mind. Keep the appetite.

